Domakonda Charanya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.03
Trividyaa Junior College	Class XII (Telangana Board of Intermediate(TGBIE) Education)	2024	97.50%
St. Mary's High School	Class X (Board of Secondary Education, Telangana)	2022	9.80%

SKILLS

Programming Languages : C, C++

Simulation & Modeling : MATLAB (incl. Simulink), TINA (circuit simulation), EViews (econometric modeling)

CAD & Embedded Systems: SolidEdge (CAD), Arduino IDE, Digital Fabrication (3D modeling & printing)

Bioinformatics & Computational Biology: NCBI databases, GEO2R, BLAST

RELEVANT COURSES

Biomedical & Core Engineering: Bioengineering, Biomedical Imaging, Basic Bioinformatics, Mathematical Modelling and Systems Biology, Medical Nanotechnology-I, Control Systems, Introduction to Embedded Systems

Mathematics: Calculus , Differential Equations, Introduction to Probability, Introduction to Statistics

Computing & Management: Introduction to Programming, Data Structures & Applications, Artificial Intelligence, Introduction to Entrepreneurship, Basic Econometrics and Forecasting

PROJECTS

Arduino-Based Sound-Tracking and Fall Detection System | Introduction to Embedded Systems (Course Project)

- Built a combined Arduino system where a sound sensor detected pitch/frequency, rotating a servo motor left or right accordingly, and ultrasonic sensors detected sudden distance changes to simulate fall detection
- Designed the circuit and coded the system in Arduino IDE, triggering an audible alert when a fall threshold was crossed

PID Controller Design for Biomedical Drug Infusion System | Control System (Course Project)

- Modeled a two-stage drug infusion system (blood concentration physiological response) as transfer functions derived from governing differential equations
- Designed and tuned a PID controller in MATLAB Simulink to achieve <5% overshoot and zero steady-state error, using the PID Tuner app
- Validated robustness by introducing actuator and sensor noise into the feedback loop; system remained stable with peak output 1.05 and only minor deviation from target

POSITIONS OF RESPONSIBILITY

Head, Campus Tours Domain, Prayas (Rural Development Cell), IIT Hyderabad | AY 2025-2026

- One of the first two Heads of the Campus Tours domain since its launch; co-designed the domain's tour scripts, guide training process, and scheduling SOPs from scratch
- Co-launched "Campus Ambassadors" with the Institute's PR Office, an initiative showcasing IITH's research to visiting parents, academics, and corporates
- Reached 4,000+ students across the domain's tours over the year, including 2,500+ through the PM SHREE outreach drive alone in a single month

Hospitality Coordinator, Inter IIT Sports Meet 2025

- Coordinated hospitality operations for visiting contingents -- accommodation, transport, catering, and welcome-desk support